CM0832 - MT5001 Elements of Set Theory
3.2 Properties of Natural Numbers

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Outline

Preliminaries

Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

The Induction Principle

Induction Principle (p. 42)

Let $P(x)$ be a property (possibly with parameters). Assume that

(a) $P(0)$ holds and

(b) for all $n \in \mathbf{N}$, $P(n)$ implies $P(n+1)$.

Then P holds for all natural numbers n.

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Proof.

The assumptions (a) and (b) simply say that the set $A = \{\, n \in \mathbf{N} \mid P(n) \,\}$ is inductive. $\mathbf{N} \subseteq A$ follows. ∎

The Induction Principle

Lemma [3.]2.1

(a) $0 \leq n$ for all $n \in \mathbf{N}$.

(b) For all $k, n \in \mathbf{N}$, $k < n + 1$ if and only if $k < n$ or $k = n$.

The Induction Principle

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Let $P(x)$ be the property "$0 \leq x$".

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(ii).

It suffices to observe that $k \in n \cup \{n\}$ if and only if $k \in n$ or $k = n$. ■

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Linear Ordering on $\mathbf{N}$

Theorem [3.]2.2

$(\mathbf{N}, <)$ is a linearly ordered set.

Linear Ordering on $\mathbf{N}$

Proof (i): $<$ is transitive on $\mathbf{N}$.

We use induction on n with $P(n)$: *"for all $k, m \in \mathbf{N}$, $k < m$ and $m < n$ imply $k < n$."*

(a) $P(0)$ holds: there is no $m \in \mathbf{N}$ with $m < 0$.

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(a) $P(0)$ holds: there is no $m \in \mathbf{N}$ with $m < 0$.

(b) $P(n)$ implies $P(n+1)$: assume $P(n)$, and let $k < m$ and $m < n+1$. By Lemma 2.1(ii), $m < n$ or $m = n$. If $m < n$, then $k < n$ by $P(n)$. If $m = n$, then $k < n$ since $k < m$. In either case $k < n + 1$ by Lemma 2.1(ii), establishing $P(n + 1)$.

∎

Linear Ordering on $\mathbf{N}$

Proof (ii): $<$ is asymmetric on $\mathbf{N}$.

Assume $n < n$ for some $n \in \mathbf{N}$. By transitivity, $n < n$ for all $n \in \mathbf{N}$. But $0 < 0$ would mean $\emptyset \in \emptyset$, which is impossible. Contradiction. ■

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Proof (iii): $<$ is a linear ordering of $\mathbf{N}$.

We prove that for all $m, n \in \mathbf{N}$, either $m < n$ or $m = n$ or $n < m$. By induction on n: the base case $P(0)$ follows immediately from Lemma 2.1(i); the inductive step uses Lemma 2.1(ii). ■

The Induction Principle, Second Version

Induction Principle, Second Version (p. 44)

Let $P(x)$ be a property (possibly with parameters). Assume that, for all $n \in \mathbf{N}$,

$$(*) \quad \text{if } P(k) \text{ holds for all } k < n, \text{ then } P(n).$$

Then P holds for all natural numbers n.

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Then P holds for all natural numbers n.

Remark

To prove $P(n)$ for all $n \in \mathbf{N}$, it suffices to prove $P(n)$ under the assumption that $P(k)$ holds for all smaller natural numbers $k < n$.

The Induction Principle, Second Version

Proof.

Assume $(*)$. Consider the property $Q(n)$: "$P(k)$ *holds for all* $k < n$."

(a) $Q(0)$ is true (there are no $k < 0$).

The Induction Principle, Second Version

Proof.

Assume $(*)$. Consider the property $Q(n)$: "*$P(k)$ holds for all $k < n$.*"

(a) $Q(0)$ is true (there are no $k < 0$).

(b) If $Q(n)$ holds, then $P(k)$ holds for all $k < n$, so $P(n)$ holds by $(*)$. Then $P(k)$ holds for all $k < n + 1$ by Lemma 2.1(ii), i.e., $Q(n + 1)$ holds.

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Proof.

Assume $(*)$. Consider the property $Q(n)$: "*$P(k)$ holds for all $k < n$.*"

(a) $Q(0)$ is true (there are no $k < 0$).

(b) If $Q(n)$ holds, then $P(k)$ holds for all $k < n$, so $P(n)$ holds by $(*)$. Then $P(k)$ holds for all $k < n + 1$ by Lemma 2.1(ii), i.e., $Q(n + 1)$ holds.

By the Induction Principle, $Q(n)$ is true for all $n \in \mathbf{N}$; since for $k \in \mathbf{N}$ there is some $n > k$ (e.g., $n = k + 1$), we have $P(k)$ true for all $k \in \mathbf{N}$. ■

Well-Ordered Sets

Definition [3.]2.3

A linear ordering $\prec$ of a set A is a **well-ordering** if every nonempty subset of A has a least element.

The ordered set $(A, \prec)$ is then called a **well-ordered set**.

Well-Ordered Sets

Theorem [3.]2.4

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Proof.

Let $X \subseteq \mathbf{N}$ be nonempty; we show X has a least element. Assume not, and consider $\mathbf{N} \setminus X$. If $k \in \mathbf{N} \setminus X$ for all $k < n$, then $n \in \mathbf{N} \setminus X$ (otherwise n would be least in X). By the second version of the Induction Principle, $\mathbf{N} \subseteq \mathbf{N} \setminus X$, so $X = \emptyset$, contradicting our assumption. ∎

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If a nonempty set of natural numbers has an upper bound in the ordering $<$, then it has a greatest element.

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Proof.

Let $A \subseteq \mathbf{N}$, $A \neq \emptyset$, have an upper bound. Let $B = \{\, k \in \mathbf{N} \mid k \text{ is an upper bound of } A \,\}$; then $B \neq \emptyset$. By Theorem 2.4, B has a least element $n = \sup(A)$.

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Let $A \subseteq \mathbf{N}$, $A \neq \emptyset$, have an upper bound. Let $B = \{\, k \in \mathbf{N} \mid k \text{ is an upper bound of } A \,\}$; then $B \neq \emptyset$. By Theorem 2.4, B has a least element $n = \sup(A)$.

It remains to show $n \in A$. Assume $n \notin A$; then $n > m$ for all $m \in A$. Since $A \neq \emptyset$, we have $n \neq 0$, so $n = k + 1$ for some $k \in \mathbf{N}$. By Lemma 2.1(ii), $k \geq m$ for all $m \in A$, so k is an upper bound of A with $k < n$, contradicting the minimality of n. ∎

Outline

References

Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on p. 3).